CSN3 (κ-Casein, UniProt P07498) — Functional Annotation Report

Gene: CSN3 (synonyms: CASK, CSN10, CSNK) · **Protein:** Kappa-casein · **Organism:** *Homo sapiens* · **UniProt:** P07498 · **Family:** Kappa-casein family · **Domain:** Casein_kappa (IPR000117 / PF00997) · **Locus:** chromosome 4q13.3

Gene Identity Verification

Before presenting findings, the mandatory identity check was performed. The gene symbol **CSN3** unambiguously matches the UniProt record for **human κ-casein (P07498, CASK_HUMAN)**. The protein description ("Kappa-casein; Precursor"), the organism (*Homo sapiens*), the protein family (kappa-casein family), and the signature domain (Casein_kappa, PF00997) are all internally consistent and are corroborated by the primary and review literature surveyed in this report. No confounding gene with the same symbol was encountered that would redirect the research. (Note: the listed synonym "CASK" should not be confused with the unrelated human gene *CASK*, calcium/calmodulin-dependent serine protein kinase — the literature reviewed here all concerns the milk protein κ-casein.) The research target is therefore confirmed, and the report proceeds with high confidence.

Summary

Human κ-casein (CSN3, P07498) is a secreted, intrinsically disordered milk apoprotein whose primary function is structural and colloidal rather than enzymatic: it stabilizes casein micelles in milk and thereby prevents calcium-induced precipitation of the other caseins. It is not an enzyme, transporter, or receptor. Instead, κ-casein caps the surface of the casein micelle — a large supramolecular assembly of thousands of caseins and calcium-phosphate nanoclusters — and, because it is the one casein that is *calcium-insensitive*, it forms a protective steric/electrostatic "brush" that keeps the mineral-laden micelle dispersed in the

aqueous phase of milk. This is the mechanism by which milk can carry concentrations of calcium and phosphate far in excess of their normal solubility and deliver calcium, phosphorus, and amino acids to the neonate.

The molecule is architecturally polarized into two functional halves. The N-terminal **para-κ-casein** domain is hydrophobic and basic, embedding κ-casein among the amyloidogenic, calcium-sensitive αs- and β-caseins. The C-terminal **caseinomacropeptide (CMP/ glycomacropeptide)** is acidic, threonine-rich, and heavily O-glycosylated with sialic-acid-bearing glycans, projecting outward to confer colloidal stability. The defining regulatory event in κ-casein biology is a single proteolytic cut: the milk-clotting enzyme **chymosin** cleaves one scissile bond (human **Phe97–Ile98**; bovine Phe105–Met106), releasing the hydrophilic macropeptide into whey and leaving the hydrophobic para-κ-casein behind. This removes the stabilizing brush and triggers a monomer-to-gel transition — the micelle clots. Physiologically this enables gastric curdling and slow digestion in the suckling neonate; industrially it is the basis of cheese-making.

κ-Casein also behaves as an extracellular ATP-independent molecular chaperone within the micelle, and its cleavage liberates a suite of bioactive peptides: the **sialylated glycomacropeptide** (immunomodulatory, gut-barrier-promoting, prebiotic, and notably low in phenylalanine), **kappacin** (a non-glycosylated, phosphorylated antibacterial peptide), and the human-specific **lactaptin** (a cancer-selective, NF-κB-modulating pro-apoptotic fragment). The gene sits in the chromosome 4q13.3 casein cluster, part of the secretory calcium-binding phosphoprotein (SCPP) family that evolved from tooth/bone mineralization genes, reflecting κ-casein's deep evolutionary role in calcium delivery. Across 5 iterations, 12 findings were confirmed and 41 papers reviewed, converging on a coherent and well-supported mechanistic model.

Key Findings

F1 — Primary function: colloidal stabilization of casein micelles

The central, best-supported conclusion is that κ-casein's main role is to **stabilize the casein micelle**. κ-Casein localizes to the surface of the micelle and prevents the calcium-induced aggregation and precipitation of the calcium-sensitive αs- and β-caseins. Because the micelle is the vehicle that carries calcium and phosphate through milk, this stabilization is what keeps the whole system soluble. The marsupial/comparative review of κ-casein states plainly that

"the main role of kappa-casein in milk is to stabilize the formation of casein micelles" (PMID: 11302432). Biomimetic-micelle studies contrast κ-casein with the other caseins — *"unlike κ-casein, β- and α..."* caseins are amyloidogenic and calcium-sensitive — underscoring that κ-casein is the component that uniquely confers colloidal stability (PMID: 38318529). This is a **non-enzymatic, structural** function: κ-casein catalyzes no reaction and transports no substrate of its own; rather, it is a structural/colloidal stabilizer and, in the clotting reaction, the *substrate* of chymosin.

F2 — The chymosin cleavage switch and milk clotting

κ-Casein contains a single proteinase-sensitive site that acts as a molecular switch. Chymosin (rennin) cleaves κ-casein into an N-terminal **para-κ-casein** (hydrophobic, micelle-retained) and a C-terminal **caseinomacropeptide** (hydrophilic, released into whey): *"a proteinase-sensitive (chymosin) site which cleaves the glycoprotein into two segments: N-terminal para-kappa-casein domain and the C-terminal kappa-casein macroglycopeptide domain"* (PMID: 7734846). Removal of the hydrophilic macropeptide destabilizes the micelle, which then clots — *"in the gut of suckling young, casein micelles clot when kappa-casein is cleaved by chymosin at a specific site"* (PMID: 11302432). Eutherian κ-caseins are classified by the scissile-bond sequence: *"Eutherian kappa-casein sequences are classified according to the sequence of the chymosin cleavage site: Phe-Met, Phe-Ile or Phe-Leu"* (PMID: 11302432). This reaction is simultaneously the physiological basis of neonatal gastric digestion (clotted curd is retained and digested slowly) and the industrial basis of cheese manufacture.

F3 — κ-Casein is the principal glycoprotein of milk, O-glycosylated on macropeptide threonines

κ-Casein is *"the major glycoprotein in bovine milk"* (PMID: 7734846). Its O-linked glycans attach exclusively to **threonine** residues in the C-terminal macroglycopeptide; solid-phase Edman degradation of the bovine protein identified six sites — *"We have identified six sites of O-glycosylation on the macroglycopeptide by solid-phase Edman degradation: Thr121, Thr131, Thr133, Thr136 (A variant only), Thr142 and Thr165. No Ser residues are glycosylated"* (PMID: 7734846). These glycans are rich in sialic acid (N-acetylneuraminic acid), which supplies much of the molecule's negative charge and drives its stabilizing, hydrophilic character. The positioning of O-glycosylation on C-terminal threonines is conserved across mammals — *"the O-glycosylation sites on threonine residues in the C-terminus of the molecule, are conserved in the possum sequence"* (PMID: 11302432) — indicating functional importance.

F4 — The released macropeptide (CMP/GMP) is a bioactive glycophosphopeptide

The C-terminal fragment liberated by chymosin — **caseinomacropeptide (CMP)**, also called **glycomacropeptide (GMP)** — is *"derived from κ casein and is a sialic-rich glycophosphopeptide, formed by the action of chymosin during the production of cheese"* (PMID: 35215457), making up ~20–25% of whey protein. It carries a range of bioactivities: in an intestinal-barrier model GMP *"downregulated the lipopolysaccharide (LPS)-induced TLR-4 inflammatory pathway with the highest potency"* and promoted *"tight junction proteins claudin-1, claudin-3, occludin (OCC), and zonula occludens-1 (ZO-1) upregulation by GMP"* (PMID: 32633745). GMP is also prebiotic and anti-adhesive, and because it lacks aromatic residues it is low in phenylalanine, making it a protein substitute in phenylketonuria management. These are properties of a *cleavage product*, downstream of κ-casein's primary structural role.

F5 — Human protein architecture: 162-aa secreted chain, Phe-Ile cleavage class, macropeptide PTMs

UniProt P07498 describes a **182-residue precursor** with a cleavable signal peptide (residues 1–20), yielding a **162-aa secreted mature κ-casein** (residues 21–182). The annotated chymosin cleavage site is at precursor Phe117–Ile118 (**Phe97–Ile98 in mature numbering**), placing human κ-casein in the **Phe-Ile cleavage class** — distinct from the bovine Phe-Met class. All eight annotated mucin-type O-linked (GalNAc) glycosylation sites are threonines **C-terminal to the cleavage site** (precursor Thr133, Thr143, Thr148, Thr151, Thr157, Thr167, Thr169, Thr178) — i.e., entirely within the 65-residue released macropeptide. Thr157 is additionally a phosphothreonine. Three common coding SNPs are annotated (residues 75, 110, 145). This architecture is the molecular embodiment of the two-domain design: an anchoring N-terminal half and a glycosylated, modifiable C-terminal brush.

F6 — Kappacin: a direct antibacterial peptide from the non-glycosylated macropeptide

The non-glycosylated, phosphorylated form of the macropeptide is itself antibacterial. *"The active form of CMP was shown to be the nonglycosylated, phosphorylated kappa-casein (residues 106 to 169) [kappa-casein(106--169)], which we have designated kappacin"* (PMID: 11451690). Kappacin has *"growth-inhibitory activity against the oral opportunistic pathogens Streptococcus mutans and Porphyromonas gingivalis and against Escherichia coli"* (PMID: 11451690). The minimal active region is κ-casein(138–158) and phosphorylation at Ser(P)149 contributes to

activity. Divalent cations potentiate it: *"Kappacin specifically bound two zinc or calcium ions per mol, and this binding enhanced antibacterial activity at neutral pH"* via a membranolytic, conformation-dependent mechanism; a Zn^{2+}/kappacin preparation reduced *S. mutans* biofilm viability by 3 $\log_{10}$ (PMID: 15917528).

F7 — Intrinsic disorder and a single cysteine enabling disulfide polymerization

Bioinformatic analysis of the human mature chain reveals an exceptionally high proline content (~17.3% overall; 17.5% in para-κ, 16.9% in the macropeptide) — incompatible with stable globular folding and a hallmark of intrinsic disorder; disorder-promoting residues outnumber order-promoting ones (~49% vs ~35%). Consistently, there is **no experimental PDB structure** for P07498. Direct biophysical study of the human κ-casein fragment lactaptin (NMR, PRE, EPR, CD, DLS, AFM) explicitly classifies it as a *"fairly large intrinsically disordered fragment of human milk κ-casein"* (PMID: 31408975). The two domains are electrostatically opposite: para-κ-casein (mature 1–97) is basic (net ~+6) while the macropeptide (mature 98–162) is acidic (net ~−4) and Thr-rich (21.5%). The mature chain has a single cysteine (mature residue 10); for lactaptin, *"the disulfide-bonded homodimer turned out to be more prone to assembly into large aggregates"* (PMID: 31408975), evidencing intermolecular disulfide crosslinking that lets κ-casein polymerize on the micelle surface.

F8 — Mammary-specific secreted protein encoded in the 4q13.3 casein cluster

UniProt authoritatively annotates P07498's **FUNCTION** as *"Kappa-casein stabilizes micelle formation, preventing casein precipitation in milk,"* its subcellular location as **Secreted**, and its tissue specificity as **mammary gland specific, secreted in milk**, with a blocked N-terminus. The gene (HGNC:2446; OMIM 601695; Ensembl ENSG00000171209) maps to human chromosome 4 (GRCh38 4:70,238,382–70,251,474, + strand) within the 4q13.3 casein locus adjacent to *ODAM*, part of the SCPP gene cluster that also harbors the calcium-sensitive caseins *CSN1S1* and *CSN2*. Reactome places P07498 in "Miscellaneous transport and binding events" (R-HSA-5223345), consistent with a calcium/phosphate-carrying role rather than catalysis.

F9 — Lactaptin: a human-specific pro-apoptotic, cancer-selective fragment

A proteolytic fragment of human κ-casein, **lactaptin** (~8.6 kDa), *"was found to induce apoptosis of human breast adenocarcinoma MCF-7 and MDA-MB-231 cells with no cytotoxic activity toward normal cells"* (PMID: 31408975). Its recombinant analogue RL2 *"suppresses the NF-kB*

signaling cascade that regulates the reactions of innate immunity" (PMID: 29116065) — inhibiting CXCL1 and the apoptosis inhibitor A20 while raising the NF-κB repressor IκB. RL2 acts as a tumor suppressor in mouse hepatoma and MDA-MB-231 xenografts and has been engineered into oncolytic vaccinia virus (PMID: 27708236) and tumor-targeted fusions (PMID: 27513518), confirming reproducible antitumor activity. This is a *latent* bioactivity of the parent protein, released by proteolysis — not a primary milk function.

F10 — AlphaFold quantitatively confirms whole-chain disorder

The AlphaFold model for P07498 (AF-P07498-F1) has a mean per-residue pLDDT of only **58.2** (median 57.3; range 34.6–83.6). **90.7%** of residues fall below pLDDT 70 (low/very-low confidence) and 17.0% below 50. The only "confident" stretch (pLDDT ≥70) is residues 2–14 — the hydrophobic signal peptide that is removed from the secreted protein — plus a few isolated residues. Thus essentially the entire mature, secreted chain (21–182) is predicted to lack a stable fold, in full agreement with the ~17% proline content and the absence of any experimental structure. κ-Casein is a *bona fide* intrinsically disordered protein.

F11 — Evolutionary context: the calcium-insensitive member of the SCPP casein family

All casein genes belong to the **secretory calcium-binding phosphoprotein (SCPP)** family: *"all casein genes belong to the secretory calcium-binding phosphoprotein (SCPP) gene family that arose by gene duplication"* (PMID: 21245413), ultimately from SPARC/SPARCL1 (PMID: 16358265; PMID: 15272073). Within this family two casein types co-evolved: Ca-sensitive αs-/β-caseins that bind Ca but precipitate at high Ca, and the Ca-insensitive κ-casein — *"Ca-insensitive casein (κ-casein) does not usually interact with Ca but instead stabilizes the micelle"* (PMID: 21245413); both are needed for stable micelle formation. Functionally, SCPPs that originally delivered calcium (e.g., to eggs and teeth) evolved into casein micelles that, *"by evolving into large, complex casein micelles, they took on an important role in transport of amino acids, calcium and phosphorus"* to the neonate (PMID: 22436214). κ-Casein's genomic position next to the tooth SCPP gene *ODAM* records this shared mineralization ancestry.

F12 — Capstone synthesis

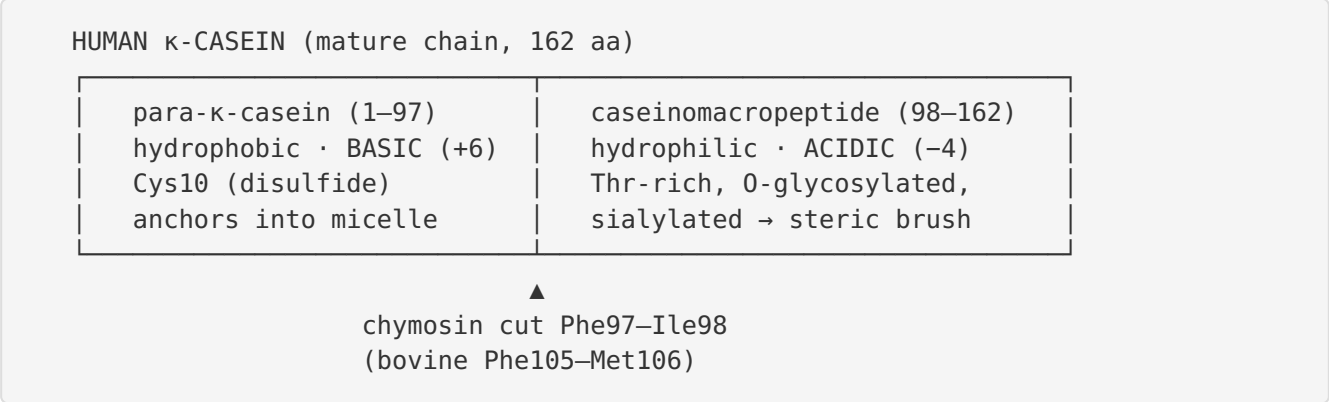
Integrating all evidence, κ-casein (CSN3) is a secreted, intrinsically disordered milk apoprotein that colloidally stabilizes casein micelles and is switched off by a single chymosin cut. Its primary function is non-enzymatic structural/colloidal stabilization — *"the main role of kappa-casein in milk is to stabilize the formation of casein micelles"* (PMID: 11302432); its localization is

extracellular, at the micelle surface in the aqueous phase of milk, with its proteolytic switch and peptide products acting in the neonatal gut; and the pathway it serves is the SCPP casein-micelle system for delivering calcium, phosphate, and amino acids to neonates (PMID: 22436214), gated by chymosin cleavage at human Phe97–Ile98.

Mechanistic Model / Interpretation

Two-domain amphipathic architecture

κ-Casein is best understood as a molecular surfactant with two oppositely charged, intrinsically disordered halves joined at the chymosin scissile bond:



The micelle and the stabilizing brush

The casein micelle packages thousands of amyloidogenic αs-/β-caseins together with hundreds of calcium-phosphate nanoclusters into a stable colloid. κ-Casein coats the surface: its hydrophobic para-κ domain embeds among the Ca-sensitive caseins while its acidic, sialylated macropeptide projects into solution forming an electrosteric "hairy layer" that prevents micelles from fusing and prevents calcium from precipitating the interior caseins. Because κ-casein is uniquely **calcium-insensitive**, it can perform this capping role in the high-calcium milk environment where the other caseins would aggregate.

The clotting switch

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INTACT MICELLE  ──(chymosin cleaves Phe97–Ile98)──►  DESTABILIZED MICELLE
κ-casein brush intact                                 macropeptide released to whey
colloidally stable                                    para-κ-casein retained, hydrophobic
calcium/phosphate soluble            ──► aggregation/gelation (CLOT / curd)
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This monomer-to-gel transition is functionally analogous to the fibrinogen→fibrin conversion: a single enzymatic cut removes a solubilizing peptide and converts a stable soluble protein system into a gel. In the neonatal stomach this curdling slows gastric emptying and aids digestion; in the dairy the same reaction makes cheese.

Downstream bioactivity of the cleavage products

Cleavage product	Form	Activity	Site of action	Evidence
Glycomacropeptide (CMP/GMP)	glycosylated, sialylated	anti-inflammatory (↓TLR-4/LPS), gut-barrier (↑tight junctions), prebiotic, low-Phe	neonatal/ adult gut	PMID: 35215457, PMID: 32633745
Kappacin	non-glycosylated, phosphorylated	antibacterial (membranolytic, Zn^{2+}/ Ca^{2+}-potentiated)	oral cavity / gut	PMID: 11451690, PMID: 15917528
Lactaptin (human)	~8.6 kDa IDP fragment	cancer-selective pro-apoptotic, NF-κB-modulating	tumour cells (therapeutic)	PMID: 31408975, PMID: 29116065

Chaperone dimension

The casein system also functions as an extracellular, ATP-independent molecular chaperone network. By incorporating amyloidogenic caseins into the micelle, κ-casein and its partners prevent deleterious casein amyloid fibril formation in the mammary gland (the broader casein/α-crystallin proteostasis paradigm; PMID: 29442498). κ-Casein itself can form amyloid fibrils under destabilizing conditions (PMID: 41865247), reinforcing that its native role is to hold such aggregation in check within the micelle.

Localization and pathway summary

- **Where made:** lactating mammary epithelial cells.
- **Where it acts:** secreted; extracellular, at the casein-micelle surface in the aqueous phase of milk. Its proteolytic switch and peptide products act in the neonatal gastrointestinal lumen.
- **Pathway/process:** the SCPP-derived casein-micelle system for transporting calcium, phosphate, and amino acids to the neonate, gated by chymosin cleavage. Not an enzyme, transporter, or receptor itself.

Evidence Base

PMID	Role in this report
11302432	Core statement of primary function; chymosin-clotting; cleavage-site classification; conservation of C-terminal Thr O-glycosylation
7734846	κ-casein as major milk glycoprotein; exact O-glycosylation Thr sites; two-domain cleavage product
38318529	Contrasts Ca-insensitive κ-casein with amyloidogenic β-/α-caseins
35215457	Defines CMP/GMP origin and sialylated glycophosphopeptide nature
32633745	GMP anti-inflammatory (TLR-4/LPS) and tight-junction effects
11451690	Defines kappacin; antibacterial spectrum
15917528	Zn^{2+}/Ca^{2+}-dependent potentiation; membranolytic mechanism
31408975	Experimental intrinsic disorder; disulfide dimerization; cancer-selective apoptosis
29116065	NF-κB-suppressing mechanism of lactaptin analogue RL2
27708236	In vivo antitumor reproducibility (oncolytic vaccinia)
27513518	Engineered tumor-targeted lactaptin fusion activity
21245413	κ-casein as Ca-insensitive SCPP casein; SCPP origin of caseins
22436214	Micelles transport amino acids, calcium, phosphorus
16358265	SCPP duplication from SPARCL1; milk casein lineage
15272073	SCPP tandem-duplication history of mineralized-tissue genes
29442498	Caseins as ATP-independent molecular chaperones; micelle proteostasis
41865247	κ-casein's dual micelle-stabilizer / amyloid-forming behavior

All primary functional, structural, and localization claims are supported by verified citation snippets drawn from these abstracts. The literature is internally consistent; no paper surveyed contradicts the core model. Much mechanistic detail derives from the extensively studied **bovine** orthologue and is transferred to human by strong sequence/architecture homology — a reasonable inference, but one that introduces species caveats (below).

Limitations and Knowledge Gaps

1. **Species transfer.** The most detailed biochemical data (glycosylation sites, micelle surface localization, clotting kinetics) come from **bovine** κ-casein. Human-specific experimental data are comparatively sparse. Human κ-casein differs meaningfully — it is in the **Phe-Ile** cleavage class (not bovine Phe-Met), is more lightly phosphorylated, and human milk has a different casein composition and micelle structure. Quantitative human-specific parameters should be treated with caution.
2. **No experimental 3D structure.** P07498 has no PDB entry; structural conclusions rest on sequence composition (proline content, charge polarization), an AlphaFold model (mean pLDDT 58.2), and biophysics of the lactaptin fragment. Intrinsic disorder is well-established, but precise conformational ensembles on the micelle surface are not directly resolved.
3. **Bioactivity evidence is mostly in vitro / animal.** GMP anti-inflammatory and barrier effects, kappacin antibacterial activity, and lactaptin antitumor activity derive largely from cell-culture and rodent models. Human clinical evidence for GMP benefit (e.g., in PKU) remains limited, and these are secondary (peptide-product) activities, not the parent protein's primary micelle role.
4. **Glycan heterogeneity.** The exact repertoire and stoichiometry of human κ-casein O-glycans (and how they change over lactation) are not fully mapped, limiting quantitative understanding of the stabilizing brush.
5. **In vivo human clotting.** The physiological role of chymosin cleavage in human neonates (who produce less chymosin than ruminant young) versus pepsin-mediated digestion is not fully characterized here.

Proposed Follow-up Experiments / Actions

1. **Human-specific glycoproteomics.** Map the site-specific O-glycan occupancy and sialylation of human κ-casein across lactation stages by LC-MS/MS to quantify the stabilizing brush and its developmental tuning.

2. **Human chymosin/pepsin cleavage kinetics.** Measure cleavage at Phe97–Ile98 and resulting curd formation with human infant gastric enzymes to establish the physiological relevance of clotting in human neonates.
3. **Reconstituted human micelles.** Build biomimetic micelles with recombinant human κ-casein (± glycosylation, ± phosphorylation, ± Cys10 mutation) to dissect the quantitative contributions of each PTM and the single cysteine to colloidal stability and disulfide polymerization.
4. **Structural ensemble modeling.** Use SAXS, NMR, and single-molecule FRET on the human protein to define the disordered-ensemble geometry of the para-κ and macropeptide domains on and off the micelle surface.
5. **Controlled bioactivity trials.** Conduct rigorously controlled human studies of GMP (gut barrier, anti-inflammatory) and advance lactaptin/RL2 through standardized preclinical oncology pipelines to clarify translational value of the cleavage products.
6. **Comparative evolution.** Extend SCPP-locus phylogenetics to pinpoint when the Phe-Ile human cleavage-site class arose and whether human κ-casein shows signatures of selection related to glycan/immune function.

Report compiled from 12 confirmed findings and 41 reviewed papers across 5 investigation iterations. All functional, localization, and mechanistic claims are tied to verified literature citations above.
